Bidding Away the Incentives to Drill

Revenue Sharing Auctions in Mexico's Oil Industry

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Revenue Sharing Auction

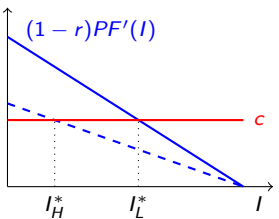
- A government leases oil fields to private companies
- Bid: each company submits a revenue share r, the share of future revenue it pays to the government
  - Companies compete on a revenue tax rate
- The company with the highest rate r wins the contract
- The winner then chooses its drilling investment I given its committed rate
  - Winner's payoff $(1 - r)\, P\, F(I)$
  - Government's payoff $r\, P\, F(I)$

Toy Model: The Winner's Investment Incentives

Revenue sharing ($100r$%)

$$\max_I (1-r)P\,F(I) - cI$$

$$(1-r)P\,F'(I^*) = c$$

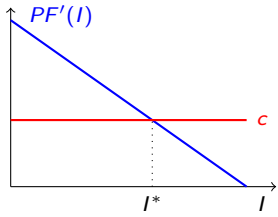


A higher r lowers investment

Fixed payment ($K)

$$\max_I \; P\,F(I) - cI - K$$

$$P\,F'(I^*) = c$$

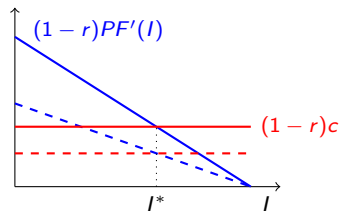


K does not enter the margin

Profit sharing ($100r$%)

$$\max_I (1-r)\big(P\,F(I) - cI\big)$$

$$P\,F'(I^*) = c$$



r does not affect investment

Solid lines: low r. Dashed lines: high r.

RQ: How Does Auction Design Affect the Winner's Drilling Decision?

- As the auctioneer, the government designs the auction rules
  - Revenue sharing auction: bidders compete on a revenue tax rate
  - Bonus auction: bidders compete on a fixed payment
  - Profit sharing auction: bidders compete on a profit tax rate
  - Minimum and maximum bids
- These design choices affect the winning bid
- Through the winner's objective function, they also affect its investment and production
- Example: revenue sharing with a minimum bid
  - The government raises the minimum bid to increase the winning bid r
  - This can discourage the winner's drilling investment I by reducing its return on investment
  - Lower investment leads to lower production $F(I)$
  - The government may lose revenue $r\,P\,F(I)$, since $r \uparrow$ but $I \downarrow$ and $F(I) \downarrow$

Contribution

Bhattacharya, Ordin, and Roberts (2022): pre-determined revenue shares

- Bonus auction with a pre-determined revenue share
- A higher revenue share delays drilling

Nguyen (2026): information disclosure in bonus auctions

- First-price bonus auction
- The winner learns about reserves from losing bids
- Disclosing losing bids encourages exploration

This paper: how revenue sharing auction design affects the winner's drilling

Why Should We Study Revenue Sharing Auctions?

Common in resource auctions in emerging markets

- Oil in Mexico and Brazil, coal in India
- Used where other auction formats are not feasible

Same logic in pay-as-produced PPA auctions

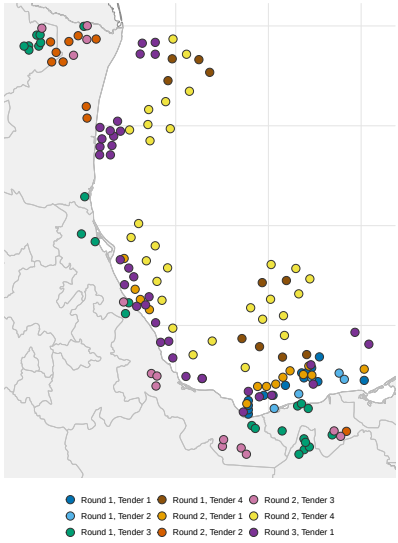
- Power plants bid an electricity price
- The lowest price wins, and the plant then chooses how much to generate at that price
- A lower price can discourage generation

Implications for auction design

- The winning bid sets the winner's incentives after the auction
- Auction design should consider post-auction incentives, not just bidding

Revenue Sharing Auctions in Mexico

- 157 blocks in 9 batches
- Firms bid a revenue share r, and the highest r wins
- r must lie between a floor and a cap (e.g. 3%–40%)
- Registered firms may stay out
- Exploration period of 5–10 years, then a production period of 20 years
- Data
  - All bids
  - Winner's monthly drilling investment and production
  - Monthly oil price
  - Government revenue



Drilling and Production

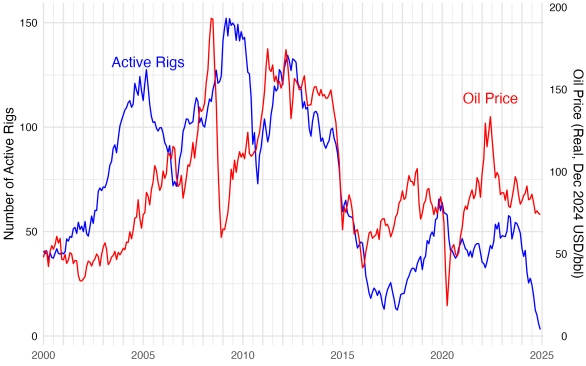
Drilling

- Boring a hole through the Earth's crust to extract oil and gas
- Oil cannot be produced without drilling
- Drilling also reveals information about the field, which is useful for neighboring fields (Kellogg, 2011)

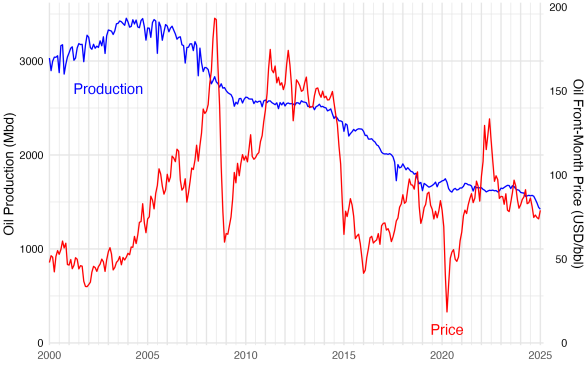
Production

- Production is constrained by drilling activity (Anderson, Kellogg, & Salant, 2018)
- Production costs almost nothing once the well is drilled

Drilling Tracks the Oil Price, but Production Does Not



Drilling



Production

How Does the Winning Bid Affect Post-Auction Outcomes?

Outcomes y_i for block i: years to drilling, exploration investment, wells drilled, whether produced, months to production, government revenue

OLS

$$y_i = \alpha + \beta\, b_i + \varepsilon_i, \qquad b_i = \text{winning bid (\%)}$$

Cox for time to drilling or production

$$h_i(\tau) = h_0(\tau)\exp(\beta\, b_i)$$

- Handles right censoring: by January 2025, 28 of 104 blocks had not drilled and 66 had not produced
- $\beta > 0$ means a shorter time to the event

IV: mean floor and cap of the other blocks in the same batch t

$$F_{-i} = \frac{1}{n_t - 1}\sum_{j \in t,\, j \neq i} F_j \qquad C_{-i} = \frac{1}{n_t - 1}\sum_{j \in t,\, j \neq i} C_j$$

- High rival floors and caps push bidders toward block i (first-stage $F = 61.9$)
- Assumes block characteristics are independent within a batch

Winning Bid and Post-Auction Outcomes

	Drilling					Production		Revenue
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Years to drilling	Years to drilling	log Expl. investment	Expl. plan-years	Wells drilled	Produced	Months to production	Government
Unit	*years*	*years*	*US$m*	*count*	*count*	*binary*	*months*	*US$m*
Estimator	*Cox PH*	*OLS*	*OLS*	*OLS*	*OLS*	*OLS*	*Cox PH*	*OLS*
Panel A. No instrument								
Winning bid (%)	-0.022***	0.088***	0.005	-0.046***	0.094	0.008***	0.023***	2.182*
	(0.005)	(0.016)	(0.006)	(0.013)	(0.063)	(0.002)	(0.007)	(1.180)
Panel B. Winning bid instrumented by the rival-auction mean floor and mean cap								
Winning bid (%)	-0.036***	0.119***	0.002	-0.057***	0.059	0.005	0.010	4.508**
	(0.013)	(0.023)	(0.009)	(0.018)	(0.087)	(0.003)	(0.014)	(2.175)
First-stage F	61.9	61.9	61.9	61.9	61.9	61.9	61.9	61.9
Total obs.	104	104	76	104	104	104	104	93
Uncensored obs.	76	–	–	–	–	–	38	–
Mean of dep. variable	1.382	3.913	4.127	4.510	6.308	0.365	26.839	28.093

A higher winning bid delays exploration but raises government revenue

Why a Structural Model?

1. A higher bid has two effects that regressions cannot separate

- Incentive: a higher share r lowers the winner's revenue and delays drilling
- Selection: a higher bid comes from a cheaper winner, who drills sooner
- The reduced-form estimates capture both
- The model separates them by recovering each winner's cost from its bid

2. Alternative auctions are not observed

- Every licence in the sample uses revenue sharing
- No data on bonus or profit sharing auctions
- The model recovers costs, bidding, and drilling, and then simulates other formats on the same blocks

Common Oil Reserves and Heterogeneous Drilling Costs

A government runs simultaneous revenue sharing auctions, one per block ℓ.

- All bidders know the block's expected reserve quantity Q_ℓ (rational expectations)
- Bidders differ in their drilling cost $C_{i\ell}$, which is private information
  - Type $Q_\ell / C_{i\ell}$ (barrels per dollar of cost) is lognormal

$$\ln(Q_\ell / C_{i\ell}) \sim N\left(m_{s(\ell)}, s^2\right), \qquad s(\ell) \in \{\text{onshore}, \text{deep water}\}$$

  - Independent across firms and blocks
- Auctions for different blocks are independent

The Auction and the Post-Auction Drilling Decision

Stage 1. Entry. A firm knows Q_ℓ, its own cost $C_{i\ell}$, the bid limits $[r_{\min}, r_{\max}]$, the entry cost c_ℓ and the current price $\bar{P}_\ell$, and decides whether to enter.

Stage 2. Bidding. Each entrant submits a revenue share r. The highest share wins.

Stage 3. Drilling and Production. Each month until the lease period ends, the winner decides whether to drill. With revenue $(1-r)P_tQ_\ell$ and n months left ($V_0 = 0$),

$$V_n(P_t; r, Q_\ell, C_{i\ell}) = \max\Big\{ \underbrace{(1-r)P_tQ_\ell - C_{i\ell}}_{\text{drill now}},\ \underbrace{\delta\, E_t\left[V_{n-1}(P_{t+1}; r, Q_\ell, C_{i\ell})\right]}_{\text{wait a month}} \Big\}$$

- Oil price $\{P_t\}$ is a random walk with $E_t[P_{t+1}] = P_t$
- Price high relative to cost: drill now. Price too low: wait a month
- Drilling costs $C_{i\ell}$ and earns the value of all future output at once

$$E_t\Big[\sum_{\tau\geq 0}\delta^\tau q_\tau P_{t+\tau}\Big] = P_t\sum_{\tau\geq 0}\delta^\tau q_\tau = P_tQ_\ell$$

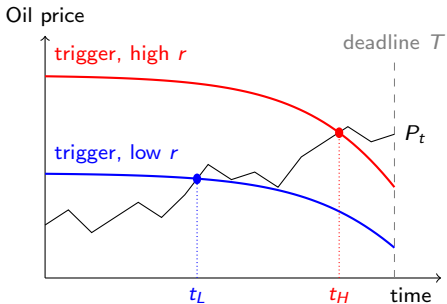
Payoffs if drilled in month t (zero otherwise): government $r\,P_tQ_\ell$, winner $(1-r)\,P_tQ_\ell - C_{i\ell}$

Proposition 1: Trigger Rule for Drilling

With n months left, the winner drills in month t iff

$$P_t \geq \underbrace{\frac{C}{Q_\ell}}_{\text{cost per barrel}} \times \underbrace{e^{y_n^*}}_{\text{option value}} \times \underbrace{\frac{1}{1-r}}_{\text{bid wedge}}$$

- Cost per barrel depends on who wins
  - A low-cost winner is more likely to drill
- Option value y^*_{T-t+1} falls as the deadline approaches
- A higher r raises the trigger, which delays or prevents drilling
  - A higher winning bid r lowers the winner's return on investment



A higher r delays drilling from t_L to t_H

Proposition 2: Optimal Bidding Anticipating Drilling

An entrant with cost C chooses its revenue share r at the contract-start price $\bar{P}_\ell$

$$\max_{r \in [r_{\min}, r_{\max}]} \underbrace{G_\ell(r)}_{\text{prob. of winning}} \quad \underbrace{V_T(\bar{P}_\ell; r, Q_\ell)}_{\text{value of the right to drill}}$$

- $G_\ell(r)$: probability that all rivals bid below r, with density $g_\ell(r)$

First-order condition

$$\underbrace{g_\ell(r)\, V_T(\bar{P}_\ell; r, Q_\ell)}_{\text{competition effect}} = \underbrace{G_\ell(r) \left(- \frac{\partial V_T(\bar{P}_\ell; r, Q_\ell)}{\partial r} \right)}_{\text{incentive effect}}$$

- Competition effect: a higher r wins more often
- Incentive effect: a higher r lowers revenue and delays drilling

Estimation: Maximum Likelihood of Bids and Drilling Years

Parameters θ to be estimated

- Mean and variance of firms' log cost efficiency $\ln(Q_\ell/C)$
- Variance of oil price shocks

Functions to be solved

- Equilibrium bid function $\rho_\ell(C_{i\ell})$
- Value function $V_n(P_t; r, Q_\ell, C_{i\ell})$

Set first, outside the likelihood

- Oil price volatility from 2010–2015 prices, discount factor δ
- Solve the value function by backward induction and compute the triggers y_n^*

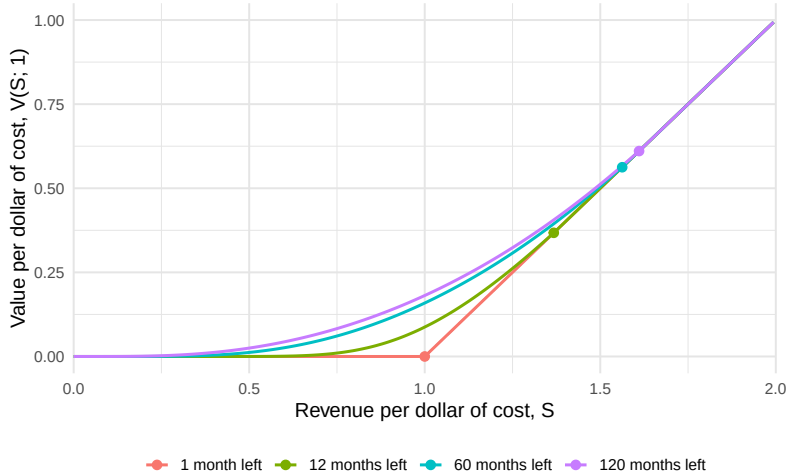
Likelihood at each trial θ

1. Bids: cost distribution $\Rightarrow$ equilibrium bid function $\Rightarrow$ probability of observed bids
2. Drilling: winner's cost (from its bid) and realized prices $\Rightarrow$ probability of its drilling year
3. Choose θ to maximize the joint likelihood of bids and drilling years

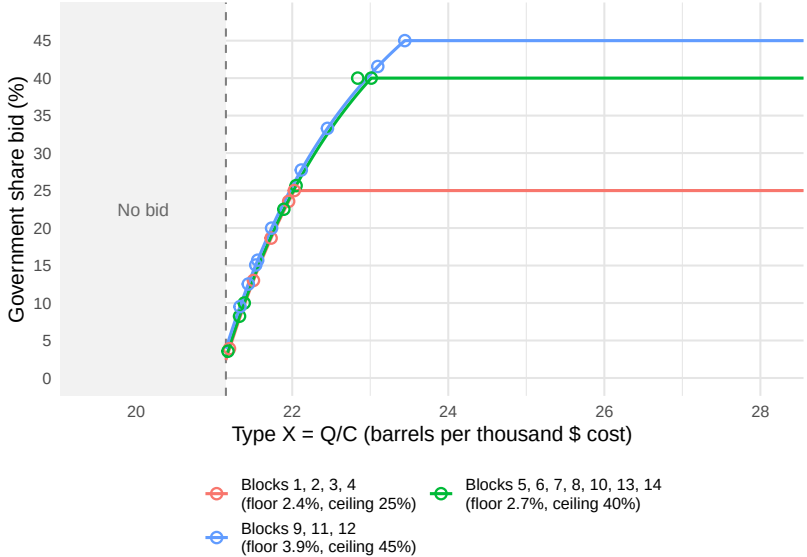
Model Fit: Predicted and Observed Outcomes

	Onshore		Deep water	
	Data	Model (SE)	Data	Model (SE)
Panel A. Entry and bidding, offered blocks				
Bids per offered block	2.67	2.67 (0.30)	1.33	1.35 (0.18)
Offered blocks with no bid	0.12	0.08 (0.05)	0.31	0.25 (0.07)
Bids at the ceiling	0.64	0.64 (0.06)	0.25	0.37 (0.07)
Mean interior share	0.16	0.21 (0.02)	0.12	0.13 (0.01)
SD of interior shares	0.10	0.11 (0.01)	0.06	0.06 (0.01)
Mean winning share	0.31	0.31 (0.01)	0.16	0.18 (0.01)
Winners at the ceiling	0.81	0.84 (0.07)	0.33	0.49 (0.08)
Panel B. Drilling, winners				
Winners that started drilling by 2024	0.57	0.77 (0.10)	0.11	0.16 (0.08)
Mean lag to the start of drilling (years)	2.33	1.89 (0.50)	5.00	3.55 (1.32)
Winners that started drilling by 2024, given actual winning share	0.57	0.78 (0.09)	0.11	0.16 (0.07)

Value of the Right to Drill



Estimated Bid Function

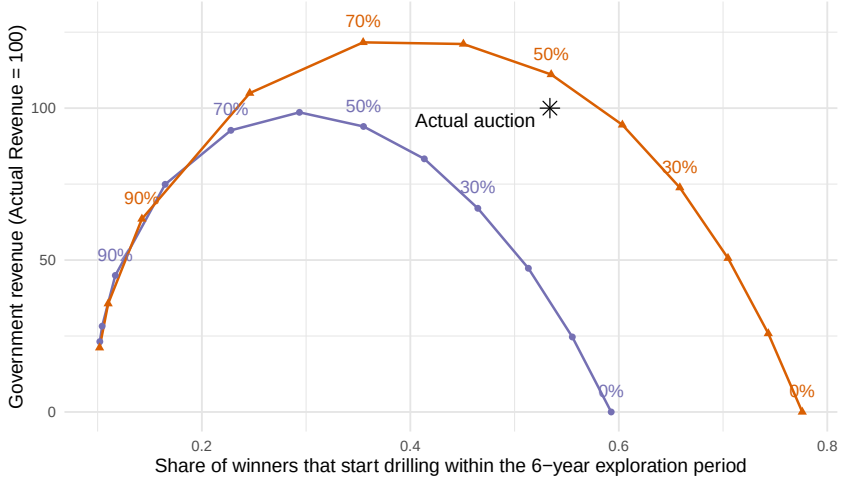


Screening Effect of Revenue Sharing Auctions

How much does the auction improve government revenue and drilling?

- The government wants to allocate a block to the lowest-cost company, but cannot observe drilling costs
- A revenue sharing auction screens for the lowest-cost company
- Alternative to the auction: a bilateral revenue-share contract
  - Worst case: a randomly chosen company leases the block at a pre-set share r
  - First best: the lowest-cost company leases the block at a pre-set share r (infeasible)
- Outcomes: government revenue and drilling probability
- Costs of less drilling
  - Less production (the estimates capture only the extensive margin)
  - Less knowledge for future exploration of nearby fields

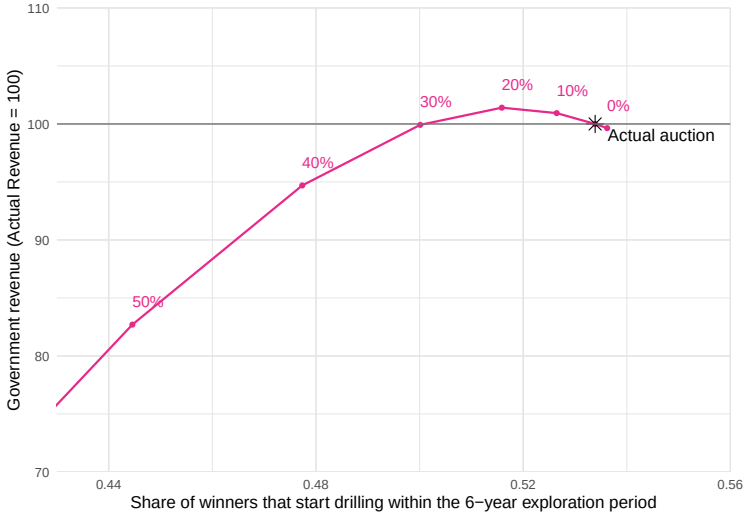
Screening Effect of Revenue Sharing Auctions (%: Revenue Share)



How a Higher Minimum Bid Changes the Auction

- A minimum bid is a common policy tool to raise the winning bid
- But it also affects the winner's drilling decision
- Government revenue $= r\,P\,F(I)$
- Raising the minimum bid leads to $r \uparrow$ but $I \downarrow$ and $F(I) \downarrow$
- The government may lose revenue if $F(I) \downarrow$ outweighs $r \uparrow$

Raising the Minimum Bid Can Reduce Government Revenue



Auction with different minimum bids (%) and no ceiling

Outcomes of Other Auction Designs

	Actual auction	Revenue share, no floor or ceiling	Bonus, first price	Profit share
Blocks awarded (%)	91.6	92.2	92.2	92.2
Winning bid	31% of revenue	48% of revenue	$40.2 million	79% of profit
Winner drills within 6 years (%)	61.2	47.6	75.8	75.8
Winner's revenue (million $)	140.5	125.6	153.0	153.0
Winner's profit (million $)	53.7	40.3	72.9	21.2
Government revenue (million $)	53.2	59.3	37.0	88.7
Total surplus (million $)	106.9	99.6	110.0	110.0

Alternative Auctions

- A revenue share distorts the contractor's drilling incentive
- A fixed payment (bonus) or a profit share does not affect the drilling decision
- But both are hard to implement in emerging markets

Bonus auction

- The winner pays a fixed amount up front, so it bears all the reserve risk
- Insurance markets are thin in emerging markets

Profit sharing auction

- The government must know the contractor's drilling cost
- Costs are hard to monitor, and the contractor has an incentive to overstate them

Conclusion

- In a revenue sharing auction, the winning bid sets the winner's drilling incentive
  - A share r raises the trigger price by $1/(1-r)$, which delays or prevents drilling
- Reduced form: a higher winning bid delays exploration but raises government revenue
- Structural model: the auction screens for the lowest-cost firm
  - But spread-out shares and higher floors delay drilling
  - Raising the floor above 20% lowers government revenue
- Alternatives keep the drilling incentive (drilling within 6 years: 76% vs. 61%)
  - Bonus auction: less government revenue ($37m vs. $53m)
  - Profit sharing auction: more government revenue ($89m vs. $53m)
  - But both are hard to implement in emerging markets

Auction design should account for the winner's incentives after the auction

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